Vision-Based Precision Pose Estimation for Autonomous Formation Flying

References: R. Punnoose, "Vision-Based Precision Pose Estimation for Autonomous Formation Flying", Stanford University, 2020.

Problem statement:
There is a leader plane and a follower plane. The follower views the leader through a camera which is gimballed at the leader (meaning the leader is always in the center of the frame). The task is to predict the "pose" of the leader plane using this image that the camera detects. A pose is the x, y, z and roll coordinates of the leader with respect to the follower.

Dataset details:
In the paper, the dataset was synthetically generated. We have also synthetically generated a dataset by building a rough mesh of an F-16, choosing 14 features on it, placing it in a "world" at coordinates x, y, and z sampled randomly, rotating it by an roll angle phi also sampled randomly, and then finding the pixel coordinates of each feature on the 2D image when being viewed from the origin.
Then we find the occluded features via ray casting, and set the (u, v) pixel coordinates to 0 as well as the is_occluded flag to 1 if it is occluded (i.e., the feature's tuple becomes (0, 0, 1) if occluded and (u, v, 0) if not occluded). The full feature vector size now becomes 14 * 3 = 42 attributes.
For the label, the paper bins the pose into one of 4800 bins. For our implementation, the true x, y, z and roll values are the labels.
One million records were generated in this manner. Train-validate-test split was 70-15-15.

Paper approach: ANN + particle filter
ANN specification:
The ANN is a multilayer perceptron with 42 inputs, two 100-unit hidden layers with ReLU activations, and a 4,800-class output. It is trained using cross-entropy loss and Adam. The saved output metrics record 40 epochs. Validation accuracy is 68.6%, test accuracy is 68.3%, and validation accuracy allowing a correct or adjacent cell is 98.9%. The adjacent-cell measure reflects the ordered pose grid and is not equivalent to exact classification accuracy.

Particle filter:
The particle filter combines the ANN pose estimates with a sampled ANN error distribution and a simplified motion model. Two saved configurations are compared: PF A uses 1,000 particles with resampling every ten cycles; PF B uses 5,000 particles with resampling every cycle. On the approach trajectory, PF A has mean position error 5.80 m and mean attitude error 4.58°; PF B has 5.31 m and 4.42°, respectively. The more frequent resampling configuration also takes longer per frame in the recorded run.

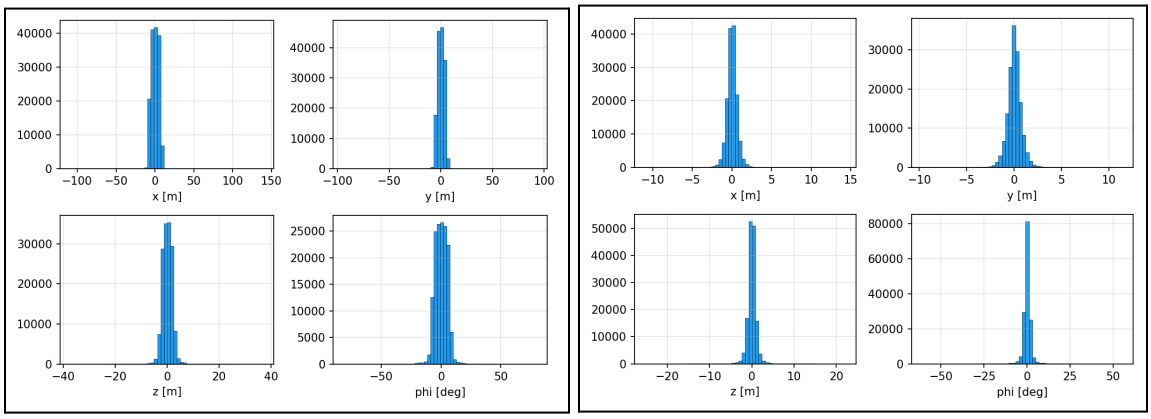
Our approach: XGBoost + Bundle Adjustment
XGBoost specification:
Four XGBoost regressors that predict continuous x, y, z, and roll values from the same 42 image features. The saved configuration uses 800 estimators, learning rate 0.05, and maximum tree depth 8. On the independent test split, the coordinate RMSEs are 0.643 m, 0.743 m, and 1.430 m for x, y, and z; roll RMSE is 2.85°. The combined three-dimensional position RMSE is 1.73 m.

Bundle adjustment:
Bundle adjustment refines XGBoost estimates by minimizing keypoint reprojection residuals, a prior around the regressed pose, and (for a multi-frame window) a second-difference smoothness term. A single-frame version uses one frame at a time. The trajectory version uses a five-frame sliding window

and a tracking initialization. Static bundle-adjustment results use 1,000 randomly selected examples from the test split.

Results

Our approach reduced the error of the paper's approach because it removed the discretization floor that came from binning.



ANN vs. XGBoost pipeline parameter error graphs

However, the bundle adjustment did not help much with the error of the XGBoost model. A different method besides particle filter or bundle adjustment alone should be explored to reduce the error of the model which predicts the pose parameters.

Pipeline configuration	Position mean (m)	Position RMSE (m)	Position max (m)	Attitude mean (°)	Attitude max (°)
ANN	5.57	5.91	11.19	3.43	8.80
ANN + PF A	5.80	6.45	12.77	4.58	8.26
ANN + PF B	5.31	5.65	9.46	4.42	7.65
XGBoost	2.53	2.73	5.25	1.87	2.87
XGBoost + BA	2.43	2.65	5.10	1.97	3.23
XGBoost + 5-frame window BA	2.42	2.63	4.99	1.97	3.40

Error comparison for the 5 different pipeline configurations tested on the follower approach scenario

Conclusions

The project demonstrates a complete simulation workflow for vision-based relative pose estimation: generate keypoint measurements, classify coarse pose with an ANN, track estimates with a particle filter, and refine continuous XGBoost predictions with reprojection-based bundle adjustment. The ANN achieves 68.6% exact-cell validation accuracy and 98.9% within-cell-neighborhood accuracy. On the recorded approach trajectory, the XGBoost and bundle-adjustment variants show lower position errors than the ANN and particle-filter baselines. Static test-subset results also show a modest improvement from single-frame bundle adjustment.